



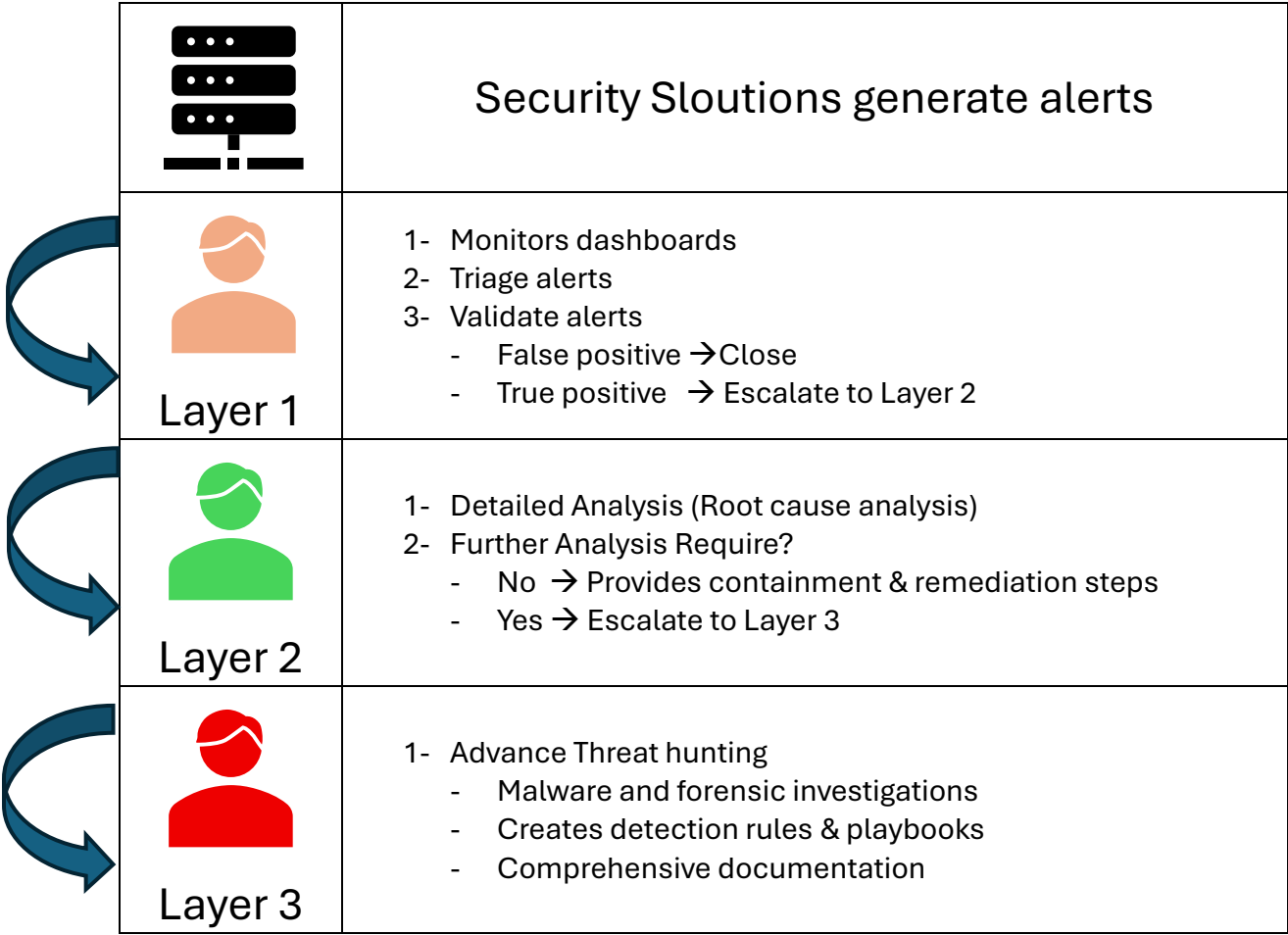
SECURITY OPERATIONS CENTRE (SOC)

Workflow Simulations

By Sultan Alangari



❖ SOC Workflow illustration



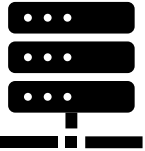


❖ Examples and simulations

- Scenario 1: Ransomware Attack incident
- Scenario 2: Lateral Movement incident
- Scenario 3: Data Exfiltration incident
- Scenario 4: Advanced Persistent Threat (APT) incident
- Scenario 5: Zero-Day Exploit incident

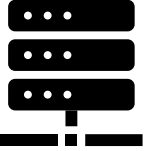


❖ Scenario 1: Ransomware Attack incident

	- Source: SIEM solution - Severity: Critical - Description: Ransomware detected on multiple endpoints - Triggered Rules: • Rule 1: Suspicious file encryption activity • Rule 2: Multiple failed login attempts followed by successful login
 Layer 1	- Receive Alert and determine Validity: • File encryption detected • Affected Endpoints: 192.x.x.1, 192.x.x.2, 192.x.x.3 - Escalated to L2
 Layer 2	- Review any recent alerts or activities related to the affected endpoints. - Findings: • Unauthorised login attempts, abnormal process activity, and data encryption. - Containment: • Isolate the affected endpoints to prevent further ransomware spread. • Inform IT and relevant stakeholders for immediate attention. - Escalated to L3
 Layer 3	- Collect and analyse memory dumps, disk images, and network traffic from the affected endpoints. - Findings: • Initial access gained via a compromised user account, followed by lateral movement and deployment of ransomware. - Recommendations: • Enhance network segmentation • Access control review

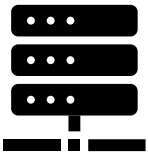


❖ Scenario 2: Lateral Movement incident

	- Source: SIEM solution - Severity: Critical - Description: Multiple failed and successful login attempts detected on various systems within a short timeframe - Triggered Rules: • Rule 1: Failed login attempts on multiple systems within 10 minutes • Rule 2: Successful logins on systems previously not accessed by the user
 Layer 1	- Receive Alert and determine Validity: • Number of failed attempts: 5 in 10 minutes • Affected systems: PC-A, PC-B, PC-C - Escalated to L2
 Layer 2	- Review any recent alerts or activities related to the affected systems. - Findings: • Source IP associated with a known APT (Advanced Persistent Threat) group. • Abnormal processes running on PC-B. • suspicious PowerShell commands executed on PC-C. - Containment: • Isolate the affected system to prevent further lateral movement. • Inform IT and relevant stakeholders for immediate attention. - Escalated to L3
 Layer 3	- Collect and analyse memory dumps, disk images, and network traffic from the affected systems. - Findings: • Evidence of a credential dumping tool used on PC-B. • Initial access gained via a spear-phishing email. - Recommendations: • Enhance network segmentation • Conduct regular phishing awareness training. • Patch vulnerabilities promptly.

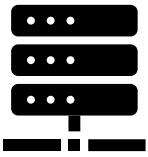


❖ Scenario 3: Data Exfiltration incident

	- Source: SIEM solution - Severity: Critical - Description: Large data transfers to an external server detected - Triggered Rules: • Rule 1: Large outbound data transfer exceeding 1GB within a short • Rule 2: Unusual traffic patterns to an unrecognised external IP
 Layer 1	- Receive Alert and determine Validity: • Large data transfer to an unrecognised external IP is suspicious - Escalated to L2
 Layer 2	- Review any recent alerts or activities related to the affected systems. - Findings: • Multiple large file accesses, network anomalies. - Containment: • Isolate the affected system to prevent further data exfiltration. • Inform IT and relevant stakeholders for immediate attention. - Escalated to L3
 Layer 3	- Collect and analyse memory dumps, disk images, and network traffic from the affected systems. - Findings: • Evidence of steganographic techniques used to hide data within images and videos. - Recommendations: • Implement data loss prevention (DLP) solutions. • Enhance monitoring for steganographic techniques.

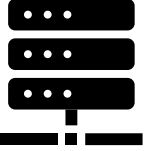


❖ Scenario 4: Advanced Persistent Threat (APT) incident

	- Source: SIEM solution - Severity: Critical - Description: A high volume of unusual DNS queries detected, indicating potential DNS tunneling activity. - Triggered Rules: • Rule 1: Excessive DNS queries from a single endpoint • Rule 2: DNS queries to domains with abnormal patterns
 Layer 1	- Receive Alert and determine Validity: • High volume of DNS queries to an abnormal domain is suspicious - Escalated to L2
 Layer 2	- Review any recent alerts or activities related to the affected systems. - Findings: • Suspicious outbound connections and abnormal process activity. - Containment: • Isolate the affected system to prevent DNS tunneling activity. • Inform IT and relevant stakeholders for immediate attention. - Escalated to L3
 Layer 3	- Collect and analyse memory dumps, disk images, and network traffic from the affected systems. - Findings: • Custom malware linked to the APT group, utilizing DNS tunneling for stealthy C2 communication. - Recommendations: • Monitoring and DNS logging • Enhance network segmentation



❖ Scenario 5: Zero-Day Exploit incident

	- Source: SIEM solution - Severity: Critical - Description: Suspicious exploit activity detected on a critical server - Triggered Rules: • Rule 1: Unusual exploit activity patterns • Rule 2: Access to critical files and system configurations
 Layer 1	- Receive Alert and determine Validity: • Exploit activity on a critical server is highly suspicious - Escalated to L2
 Layer 2	- Review any recent alerts or activities related to the affected systems. - Findings: • Abnormal processes running and unauthorised changes to system configurations. - Containment: • Isolate the affected server to prevent further exploit activity. • Inform IT and relevant stakeholders for immediate attention. - Escalated to L3
 Layer 3	- Collect and analyse memory dumps, disk images, and network traffic from the affected systems. - Findings: • Unpatched software with a known vulnerability, no available patch yet. - Recommendations: • Work with the software vendor to expedite a patch. • Enhance network monitoring.